

Supplementary methods

Preparation of $(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$ solid solution.

$(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$ solid solutions, samples A, B, and C, were prepared by heating a mixture of Ga_2O_3 and ZnO powders (ca. 2 g) under NH_3 flow ($250 \text{ mL} \cdot \text{min}^{-1}$). Detailed preparation conditions are summarized in Table 1 (see below). After nitridation, the sample was cooled to room temperature under NH_3 flow. Powder X-ray diffraction and energy-dispersive X-ray analyses and diffuse reflectance spectroscopy confirmed the obtained samples A, B, and C to be solid solutions of GaN and ZnO with $x = 0.12$, 0.16 , and 0.42 (band gap of 2.68, 2.60, and 2.43 eV), respectively.

Supplementary Table 1. Preparation conditions of $(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$ solid solutions.

Sample	Amount of starting materials		Nitridation conditions	
	Ga_2O_3 / g	ZnO / g	Temp. / °C	Time / h
A	1.08 ^a	0.94 ^c	950	0.25
B	1.08 ^a	0.94 ^d	850	15
C	0.88 ^b	1.15 ^d	950	1.5

^a Purchased from High Purity Chemicals, Purity 99.9%.

^b Prepared by hydrolysis of $\text{Ga}(\text{NO}_3)_3 \cdot x\text{H}_2\text{O}$ in the presence of urea, followed by heating in air at 500 °C for 2 h.

^c Purchased from Wako Pure Chemicals, Purity 95%.

^d purchased from Kanto Chemicals, Purity 99%.

Modification of $(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$ with Rh-Cr mixed-oxide nanoparticles.

$(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$ powder (0.3–0.4 g) was added to distilled water (3–4 mL)

containing an appropriate amount of $\text{Na}_3\text{RhCl}_6 \cdot 2\text{H}_2\text{O}$ (Kanto Chemicals, 97% as Rh) and $\text{Cr}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (Wako Pure Chemicals, 99.9%) in an evaporating dish in a water bath. The suspension was evaporated under constant stirring with a glass rod. The resulting powder was collected and was heated in air at 350 °C for 1 h.

Photocatalytic reactions and measurement of quantum efficiency.

Overall water splitting reactions for the measurement of the quantum efficiency (see Figure 1a) were carried out in a Pyrex top-irradiation-type reaction vessel connected to a glass closed gas circulation system. The photocatalyst powder (0.1 g) was dispersed in an aqueous solution adjusted to pH 4.5 by H_2SO_4 (100 mL), and the reactant solution was evacuated several times to remove air completely. The solution was then irradiated using a 300 W xenon lamp attached with a cutoff filter.¹¹

For the reaction shown in Figure 1b, a Pyrex inner-irradiation-type reaction vessel and a 450 W high-pressure mercury lamp were used. The amount of photocatalyst used for the reaction was 0.3 g. The reactant solution was an aqueous solution adjusted to pH 4.5 by H_2SO_4 (370 mL). UV light via a 450 W high-pressure mercury lamp was blocked by a NaNO_2 aqueous solution (2 M) filled in Pyrex glass tube. In this case, it is considered that an emission of a high pressure mercury lamp at 436 nm mainly contributes for the reaction. The evolved gas was analyzed by gas chromatography.

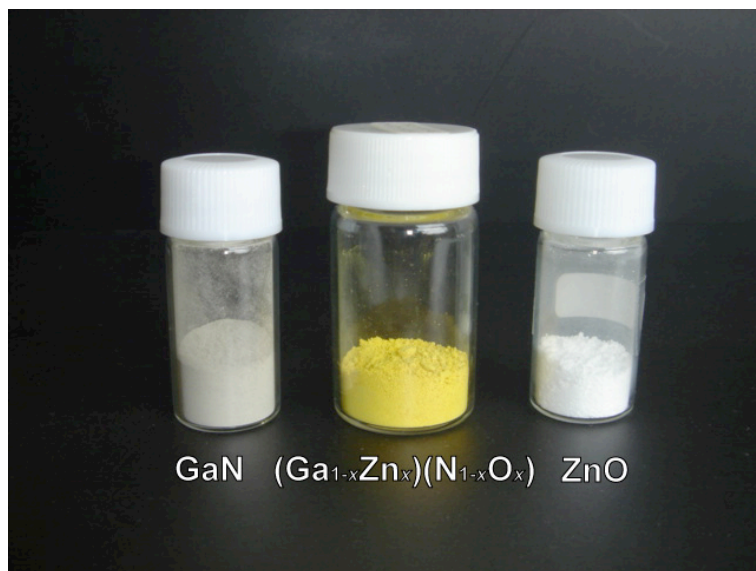
The quantum efficiency (Φ) was calculated using the following equation: $\Phi (\%) = (2 \times R/I) \times 100$, where R and I represent the number of evolved H_2 molecules and the number of incident photons, respectively. Here, Φ is quantum efficiency or photonic efficiency (ξ) proposed by Serpone *et al.*¹, where it is assumed that all incident photons are absorbed by the photocatalyst. The number of incident photons was measured by using a calibrated Si photodiode. The quantum efficiencies plotted in the Figure 1a were estimated

by using several cut-off filters, and were calculated on the basis of the number of the incident photons in each wavelength region. Accordingly, each value is plotted in the middle of two cut-off wavelengths, and the quantum efficiencies given in the Figure 1a indicate the average ones between two cut-off wavelengths. A band pass filter at 400 nm was also applied to the measurement of the quantum efficiency, and the thus-estimated quantum efficiency was 3.0 ± 0.2 %, which was in reasonable agreement with the results of Figure 1a.

Supplementary reference

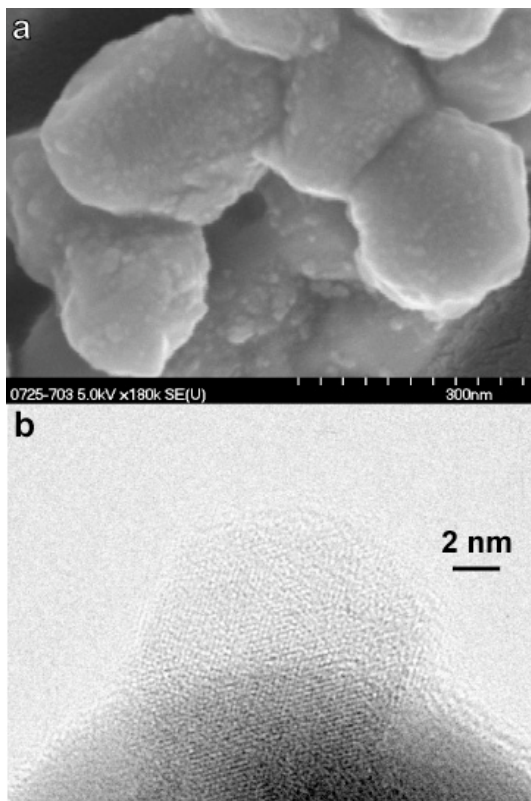
1. Serpone, N. *J. Photochem. Photobio. A: Chem.* **104**, 1–12 (1997).

Supplementary Figure S1



Photograph of the $(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$ solid solution (sample A, see supplementary method) showing GaN and ZnO.

Supplementary Figure S2



a, Scanning electron microscopy and **b**, transmission electron microscopy images of $(\text{Ga}_{1-x}\text{Zn}_x)(\text{N}_{1-x}\text{O}_x)$ (sample B, see supplementary method) containing Rh-Cr mixed-oxide nanoparticles.